

Foraging and diet

Many bats eat insects: some species forage for them among shrubs and trees, while others skim the surface of the forest canopy to catch higher-flying insects. A single bat may eat hundreds of mosquitoes in one night (thereby lessening the incidence of malaria in other animals). Other bats eat fruit, and some use their long tongue to feed on pollen and nectar. Vampire bats use their sharp teeth to make a small incision in the skin of an animal while it sleeps and then drink the blood. Carnivorous bats prey on lizards and frogs; fish-eating bats use the hooked claws on their powerful feet to capture fish (see below).



FEEDING ON FRUIT

Most fruit-eating bats use their senses of sight and smell to find food and so lack facial ornaments used in echolocation. This epauletted fruit bat is eating a mango: a group of these bats in a plantation can cause considerable damage. Since fruit bats require a constant supply of ripe fruit, they are found mostly in tropical areas. Fruit bats often feed in groups and fly long distances in search of food.

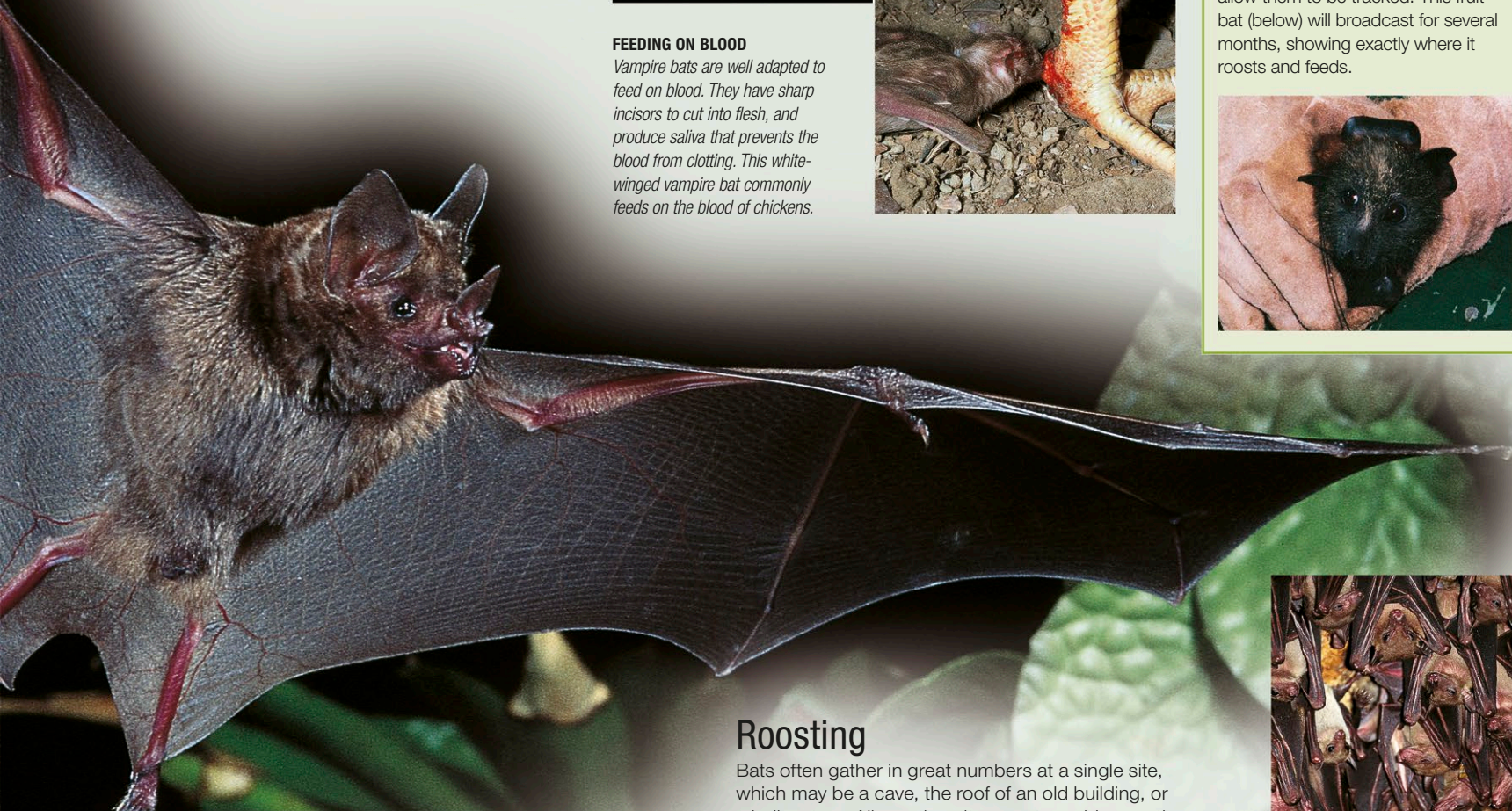


FEEDING ON BLOOD

Vampire bats are well adapted to feed on blood. They have sharp incisors to cut into flesh, and produce saliva that prevents the blood from clotting. This white-winged vampire bat commonly feeds on the blood of chickens.

CONSERVATION

Despite being rich in terms of species, bats face a huge variety of threats. Globally, one of the most serious is habitat loss, particularly in forested regions. Fruit-eating bats are also persecuted as pests, and are killed by colliding with power lines, while insect-eating bats are threatened by white nose disease—a potentially deadly fungal pathogen that first came to light in North America in 2006. Bats are now routinely fit with microchips and radio collars that allow them to be tracked. This fruit bat (below) will broadcast for several months, showing exactly where it roosts and feeds.



FLYING MAMMAL

Bats, the only mammals that can fly, have structural adaptations that allow them to make up-and-down movements of their wings (in the same way as birds). The open wings of this New World leaf-nosed bat reveal the extent of the wing membranes.

Roosting

Bats often gather in great numbers at a single site, which may be a cave, the roof of an old building, or a hollow tree. All roosting sites must provide a resting place that offers protection from predators, the heat of the sun, the low temperatures of winter (hibernating roosts), and rain. Bamboo bats are small enough to roost in the hollow stems of a plant, while some species of leaf-nosed bats bite into leaf stems so that the leaf droops downward, forming a tent around them. Why bats gather in such large numbers is not fully understood; however, at the end of hibernation, bats living in colonies often weigh more than species that do not.



A PLACE TO REST

Bats, such as these fruit bats, commonly roost in caves during the day, emerging at dusk to feed. Some species use the same roost for many years and gather in groups of many thousands.



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LANDING

As the bat attaches itself to a tree (upside down), it continues to hold the fish firmly in its mouth.



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DEVOURING THE PREY

The fish is eaten head first. The bat may use its wings to manipulate the food.



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ROOM FOR PLENTY

The bat has highly elastic cheeks, which can be extended during feeding.



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ALMOST GONE

With its meal nearly finished, the fisherman bat will soon begin the hunt for more food.